NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

MS4672 – Week 5

High Throughput Experimental Methods for Materials Discovery

Emha Bayu Miftahullatif (EMHA0001@e.ntu.edu.sg)
Pethe Shreyas Dinesh (SHREYASD002@e.ntu.edu.sg)

Also thanks to:
Dr. Danny Ren Zekun

School of Materials Science and Engineering

Today's class

- 1st half : Introduction to High-Throughput Experimentation (120 mins)
  - Introduction to High Throughput experimentation (20 mins)
  - Opentrons automated pipetting robots + Code walkthrough (40 mins)
  - In-class project: automated solution mixing (60 mins)

- 2nd half: OpenCV image manipulation (60 mins)
  - Introduction to OpenCV (30 mins)
  - Code walkthrough (20 mins)
  - Homework (10 mins)

1st half
Introduction to High-Throughput Experimentation

Conventional Discovery

① Serendipity-driven

Gorilla glass (1952)

Pure luck

② Persistent trial and error

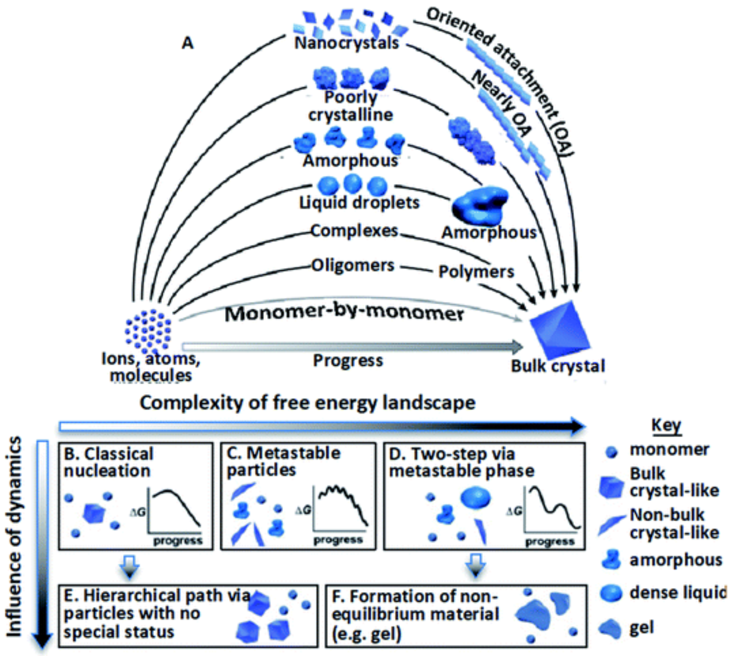
Light bulb filament (1880)

Hypothesis-driven

Slow and expensive

Why?

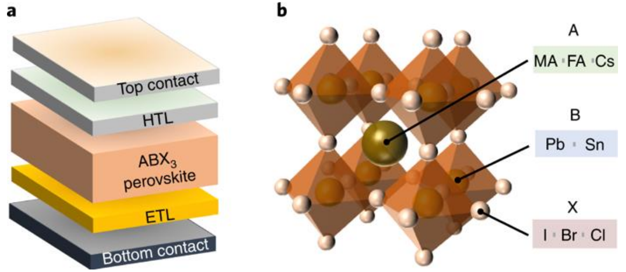
Because materials synthesis is complex



Faraday Discuss., **2022**, 235, 9

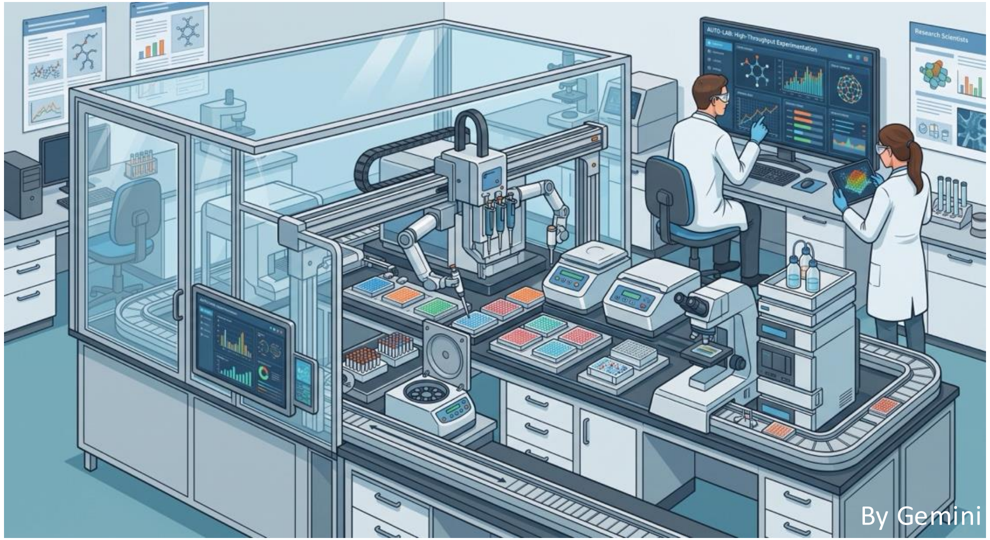
Even more complex to make device out of it

solar cells



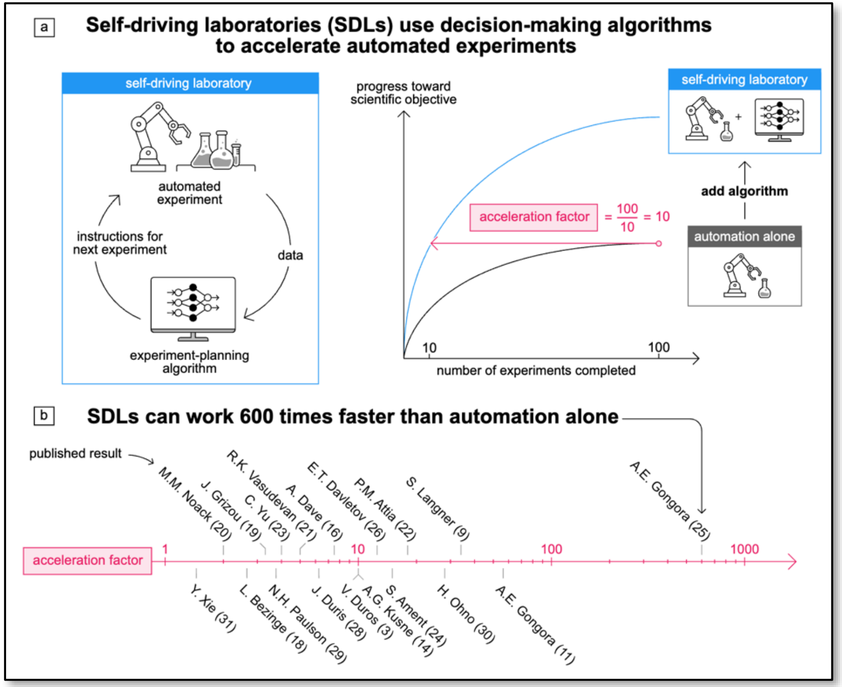
The new mainstream:

High Throughput Experimentation + self-driving lab



- Accelerated data acquisition
- Enhanced Reproducibility
- Resource and Cost Efficiency
- AI-Driven Optimization
- Continuous Operation
- Broad Parameter Mapping
- Seamless Data Integration

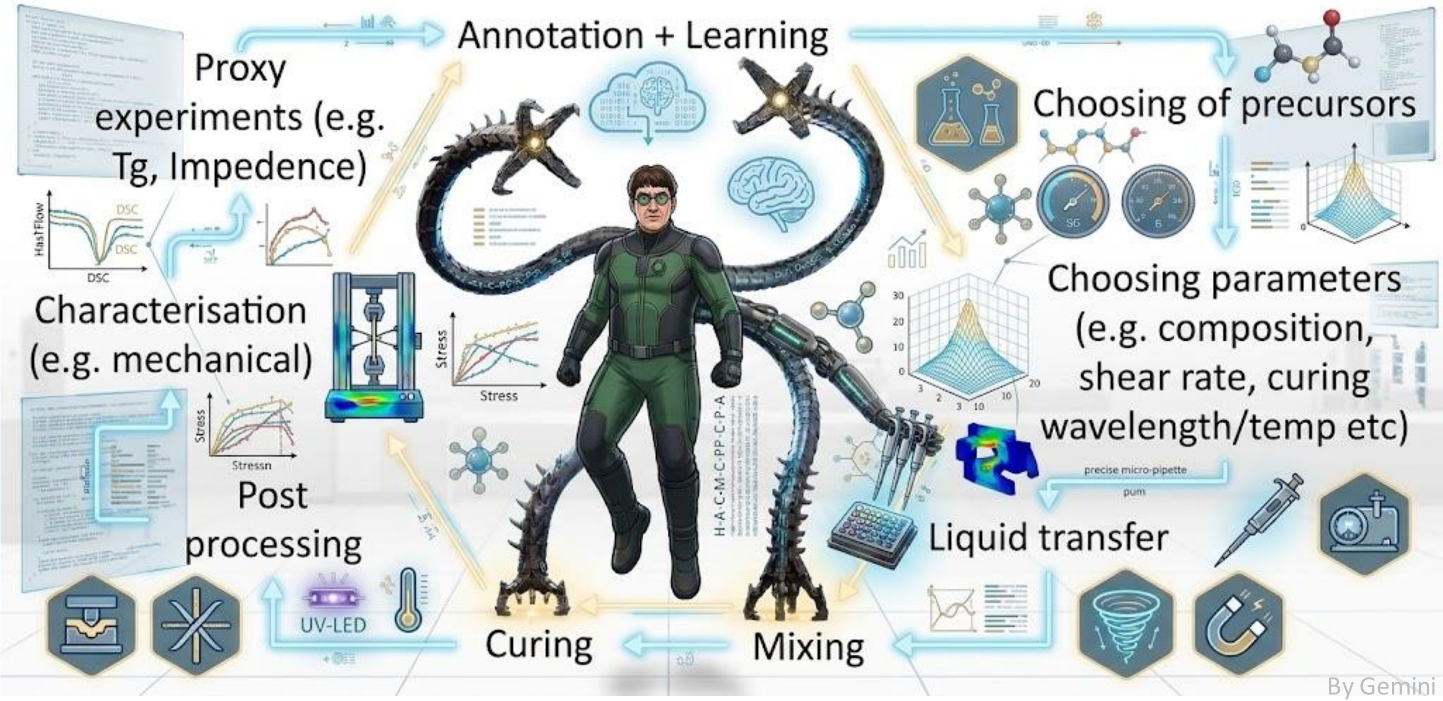
The new mainstream:
High Throughput Experimentation + self-driving lab



MRS Bulletin, **2023**, 48, 173–178.

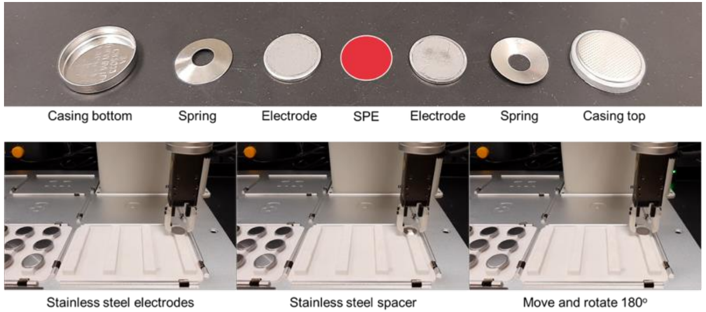
High Throughput Experimentation

What tasks need to be done?

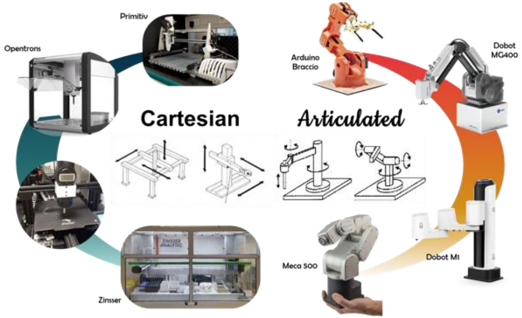


Types of modules to accomplish the tasks

Modules for making things out of stuff

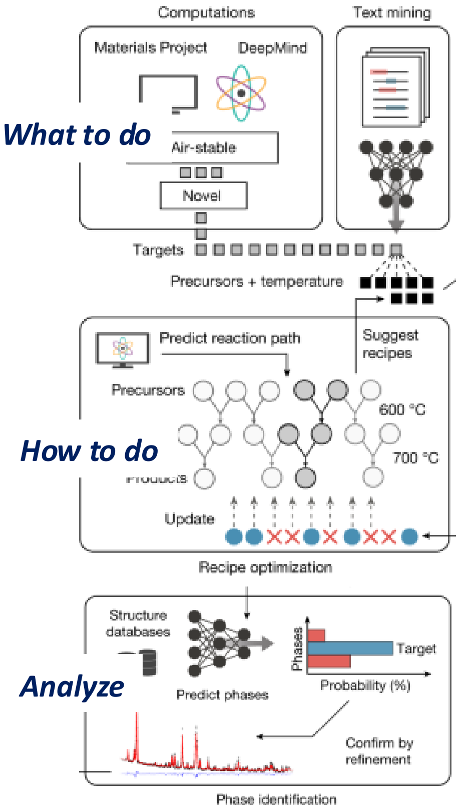


Modules for moving stuff



Modules for Testing

Decision-making brain



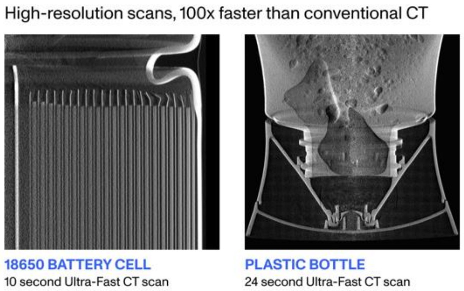
Types of experimental techniques in HTE

- Point-wise experiments (e.g. mechanical, spectrum analysis, electrical etc…)
- Parallel/Simultaneous experiments (e.g. impedance spectroscopy, gas adsorption etc…)
- Proxy experiments (Optical, electrical measurements)
- Ultrafast analysis

Nanoindenter

Potentiostat

Hyperspectrum

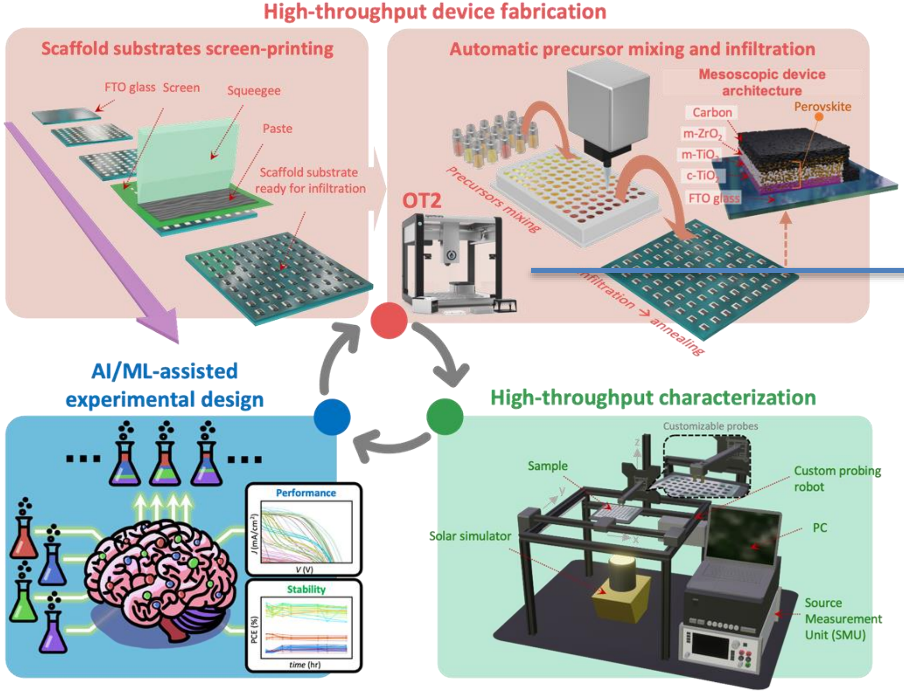


Ultrafast CT

https://www.bruker.com/en/products-and-solutions/test-and-measurement/nanomechanical-test-systems/hysitron-ti-980-nanoindenter.html
https://www.biologic.net/product_category/potentiostats-galvanostats/
https://www.lumafield.com/product-feature/ultra-fast-ct-scanning
Wikipedia

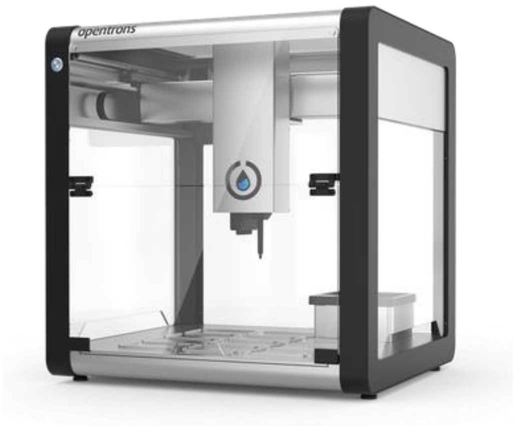
Workflow example

Optimizing perovskite precursors for printable solar cells



Next:

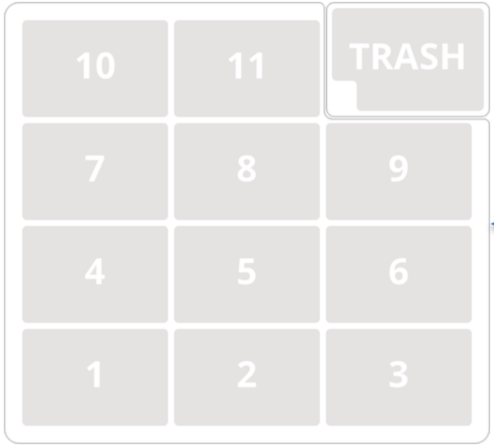
We will learn to use this

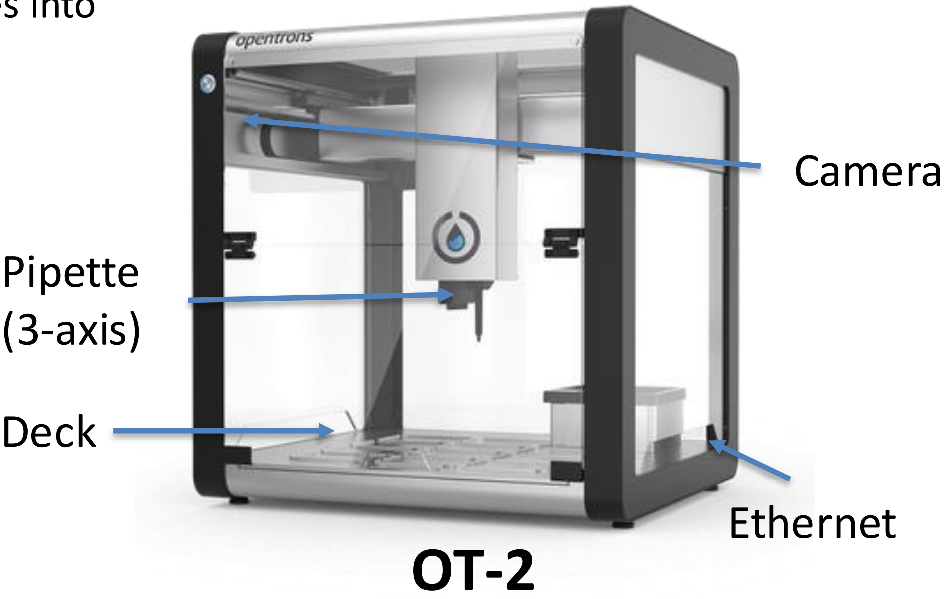


Miftahullatif, Emha B.; et al. Machine-Learning-Driven in-Device Optimization of All-Printed Perovskite Solar Cells. *ACS Energy Lett.* **2025**, *10* (8), 3952–3961.

Introduction to Opentrons

- 3-axis Liquid handling robot
- Picks liquid from reservoir (resource), pipettes into receptacle, disposes tips in trash
- Python controlled
- Standardized labware





Introduction to Opentrons

Modules

Pipettes

thermocycles

Heater

Heater-shaker

Labwares

Protocol

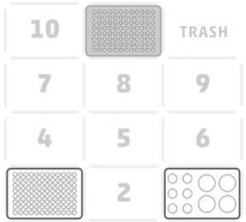
protocol_modified.py

protocol_modified.py > ...

```python
from opentrons import protocol_api

# Protocol metadata - required for all protocols
metadata = {
    'protocolName': 'Basic Liquid Handling',
    'author': 'Student Name',
    'description': 'Simple protocol for learning OT-2 operation',
    'apiLevel': '2.13'
}

def run(protocol: protocol_api.ProtocolContext):
    """
    This function defines what the robot will do.
    It's called automatically when the protocol runs.
    """

    # 1. Load labware (define what equipment is on the deck)
    # Slot numbers 1-11 refer to positions on the robot deck
    plate = protocol.load_labware('corning_96_wellplate_360ul_flat', '1')
    tiprack = protocol.load_labware('opentrons_96_tiprack_300ul', '11')
    reservoir = protocol.load_labware('opentrons_10_tuberack_falcon_4x50ml_6x15ml_conical', '3')

    # 2. Load pipette (define which pipette to use)
    # 'right' or 'left' refers to the pipette mount position
    pipette = protocol.load_instrument(
        'p300_single_gen2',
        'left',
        tip_racks=[tiprack]
    )

    # 3. Protocol steps (define the liquid handling operations)

    # Pick up a tip
    #pipette.pick_up_tip()
    pipette.pick_up_tip(tiprack['A1'])

    # Aspirate 20µL from reservoir well A1
    pipette.aspirate(20, reservoir['A3'])

    # Dispense 20µL to plate well A1
    pipette.dispense(20, plate['A1'])

    # Repeat for multiple wells
    for well in ['A2', 'A3', 'A4', 'A5', 'A6']:
        pipette.aspirate(20, reservoir['A3'])
        pipette.dispense(20, plate[well])

    # Drop the tip when finished
    pipette.drop_tip()
```

Protocol

```
protocol_modified.py ×
protocol_modified.py > ...
from opentrons import protocol_api

# Protocol metadata - required for all protocols
metadata = {
    'protocolName': 'Basic Liquid Handling',
    'author': 'Student Name',
    'description': 'Simple protocol for learning OT-2 operation',
    'apiLevel': '2.13'
}

def run(protocol: protocol_api.ProtocolContext):
    """
    This function defines what the robot will do.
    It's called automatically when the protocol runs.
    """

    # 1. Load labware (define what equipment is on the deck)
    # Slot numbers 1-11 refer to positions on the robot deck
    plate = protocol.load_labware('corning_96_wellplate_360ul_flat', '1')
    tiprack = protocol.load_labware('opentrons_96_tiprack_300ul', '11')
    reservoir = protocol.load_labware('opentrons_10_tuberack_falcon_4x50ml_6x15ml_conical', '3')

    # 2. Load pipette (define which pipette to use)
    # 'right' or 'left' refers to the pipette mount position
    pipette = protocol.load_instrument(
        'p300_single_gen2',
        'left',
        tip_racks=[tiprack]
    )

    # 3. Protocol steps (define the liquid handling operations)

    # Pick up a tip
    #pipette.pick_up_tip()
    pipette.pick_up_tip(tiprack['A1'])

    # Aspirate 20µL from reservoir well A1
    pipette.aspirate(20, reservoir['A3'])

    # Dispense 20µL to plate well A1
    pipette.dispense(20, plate['A1'])

    # Repeat for multiple wells
    for well in ['A2', 'A3', 'A4', 'A5', 'A6']:
        pipette.aspirate(20, reservoir['A3'])
        pipette.dispense(20, plate[well])

    # Drop the tip when finished
    pipette.drop_tip()
```



Everything within run() will be executed

Define and load the labwares

Define and load the modules

Protocol steps

Ref: https://docs.opentrons.com/python-api/

Core Workflow Overview of OT2

① **Check Health**
Verify robot connectivity and status.

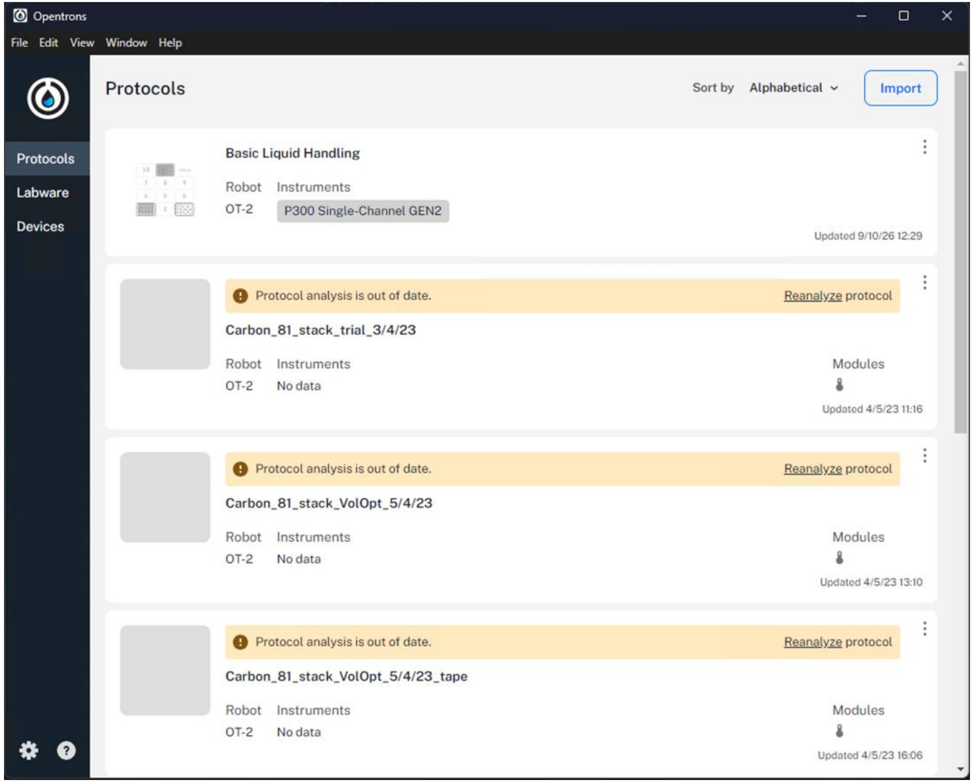
→

② **Upload Protocol**
Send your Python protocol file to the robot.

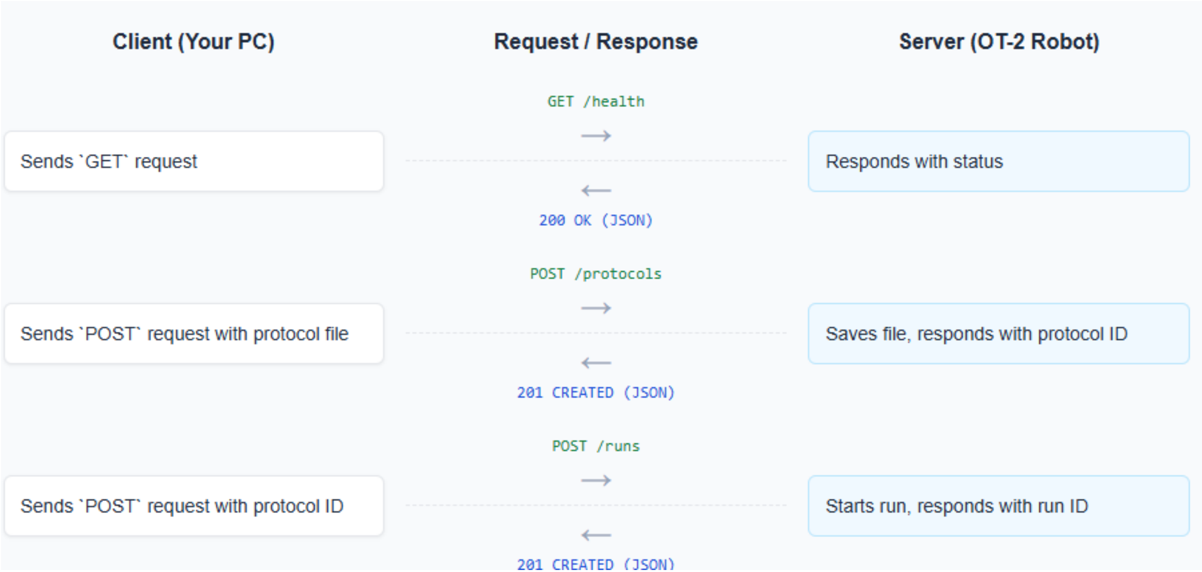
→

③ **Execute Run**
Start the experiment and monitor its progress.

Manual run via Apps

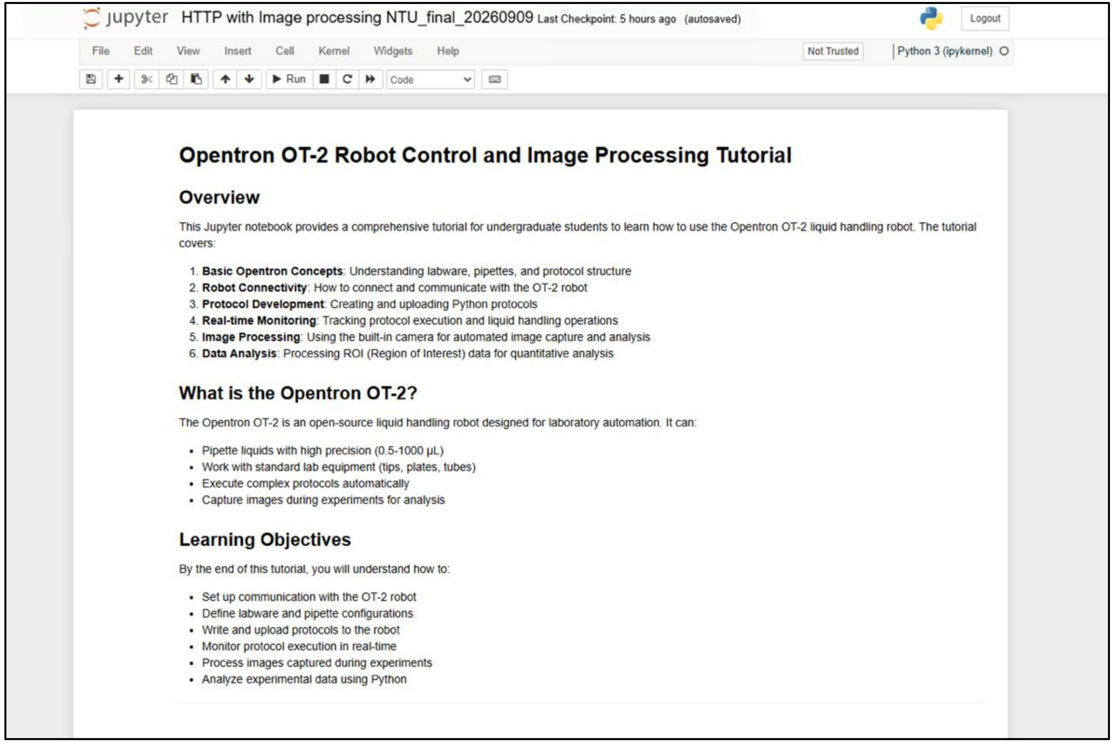


Communication protocol using Rest API



- The interaction between your computer (the client) and the OT-2 (the server) is managed through a REST API.
- Scripts sends HTTP requests to specific URLs on the robot, and the robot responds with JSON-formatted data.

Automated run via RestAPI--Code walkthrough



jupyter HTTP with Image processing NTU_final_20260909 Last Checkpoint: 5 hours ago (autosaved)

Logout

File Edit View Insert Cell Kernel Widgets Help

Not Trusted

Python 3 (ipykernel)

Run Code

Opentron OT-2 Robot Control and Image Processing Tutorial

Overview

This Jupyter notebook provides a comprehensive tutorial for undergraduate students to learn how to use the Opentron OT-2 liquid handling robot. The tutorial covers:

1. **Basic Opentron Concepts**: Understanding labware, pipettes, and protocol structure
2. **Robot Connectivity**: How to connect and communicate with the OT-2 robot
3. **Protocol Development**: Creating and uploading Python protocols
4. **Real-time Monitoring**: Tracking protocol execution and liquid handling operations
5. **Image Processing**: Using the built-in camera for automated image capture and analysis
6. **Data Analysis**: Processing ROI (Region of Interest) data for quantitative analysis

What is the Opentron OT-2?

The Opentron OT-2 is an open-source liquid handling robot designed for laboratory automation. It can:

- Pipette liquids with high precision (0.5-1000 µL)
- Work with standard lab equipment (tips, plates, tubes)
- Execute complex protocols automatically
- Capture images during experiments for analysis

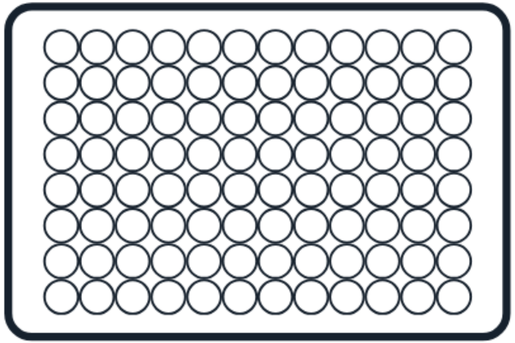
Learning Objectives

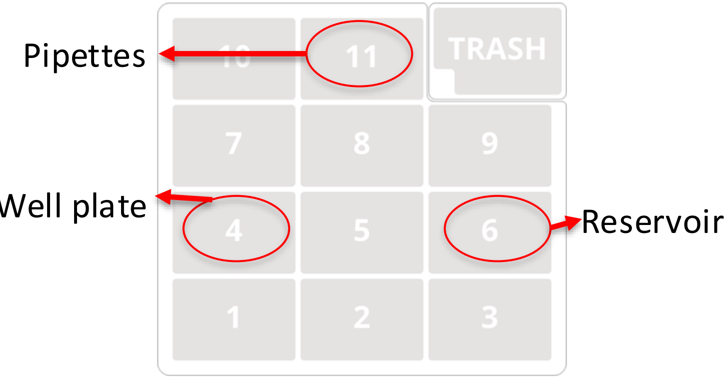
By the end of this tutorial, you will understand how to:

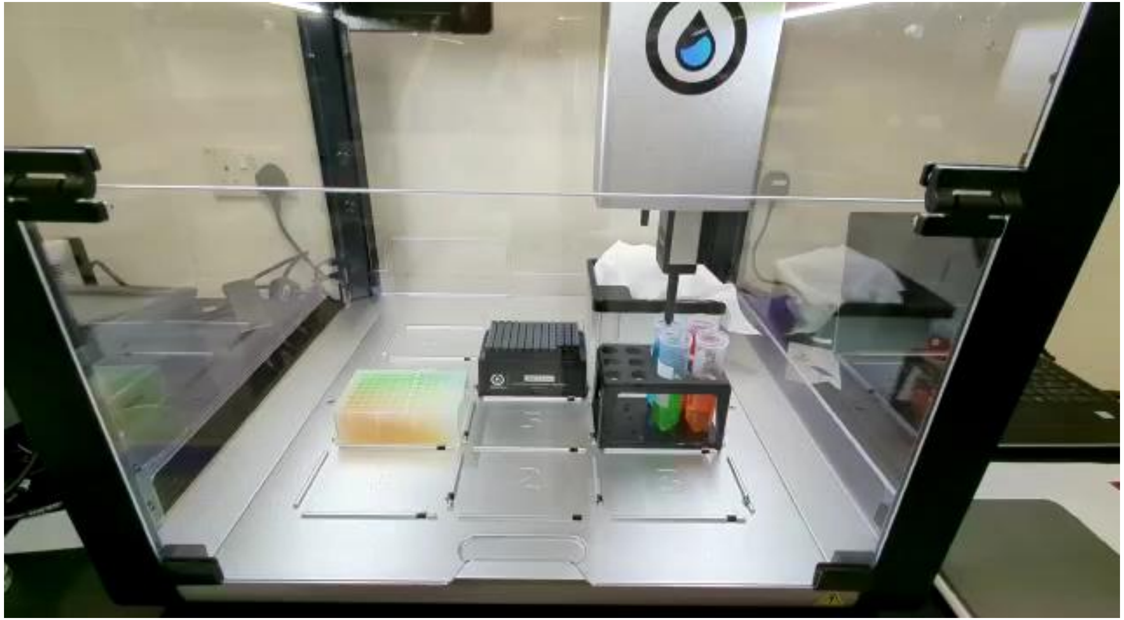
- Set up communication with the OT-2 robot
- Define labware and pipette configurations
- Write and upload protocols to the robot
- Monitor protocol execution in real-time
- Process images captured during experiments
- Analyze experimental data using Python

https://github.com/MH-BY/MS4672/

Automated colour mixing using Opentrons





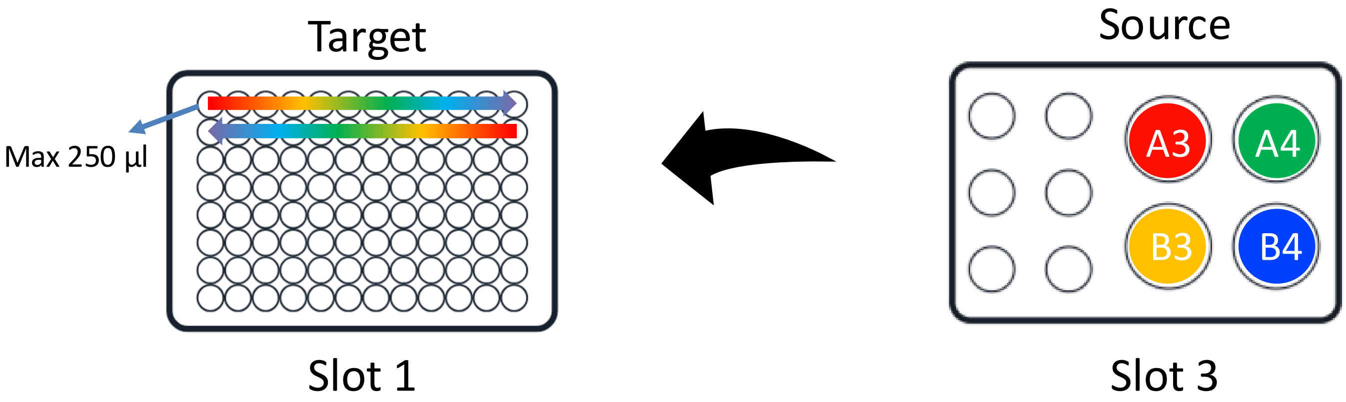


- Amount of colour x transferred to each well
- Efficiency in liquid transferring

In-class Project (45mins coding + 15mins run)

(Follow the code provided to achieve automated color mixing)

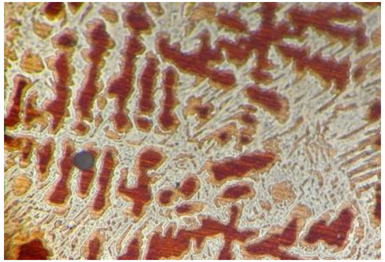
1. Modify the code provided (protocol_modified.py) to achieve automated rainbow color mixing in two rows.



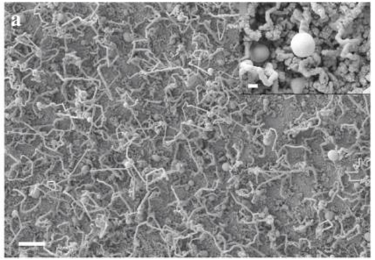
2. Name it <matric_no>.py (e.g. G2201923G.py). Submit your code to: https://github.com/MH-BY/MS4672/tree/main/week-5/Proj-submission

2nd half
OpenCV image manipulation

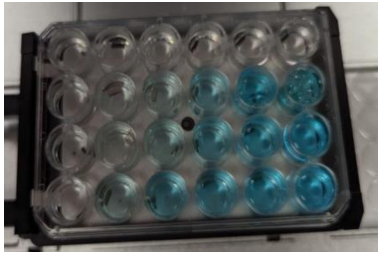
A lot of materials data is visual



Optical microscopy

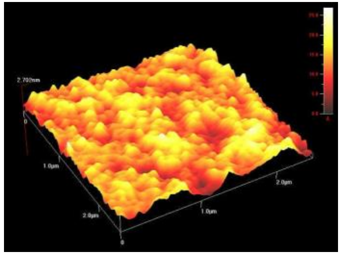


SEM

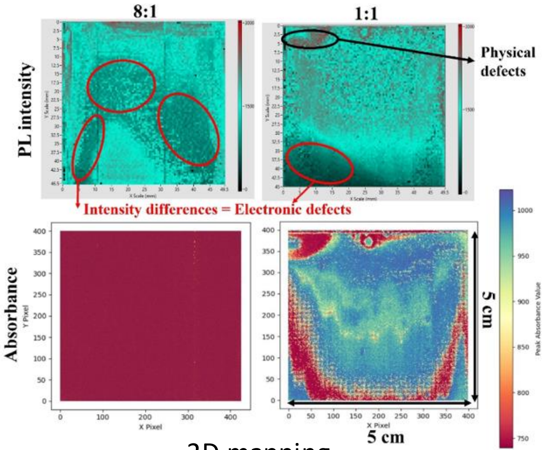


Photos of the samples

- Morphology
- Composition
- Homogenity
- Concentration
- Hardness
- Particle size
- Spectral Absorption



AFM



2D mapping

Image analysis software – Fiji, Gwiddiyon, Ilastik

https://science-saved.com/afm
https://ieeexplore.ieee.org/abstract/document/10343888
https://chemistry-europe.onlinelibrary.wiley.com/doi/epdf/10.1002/cssc.202402499

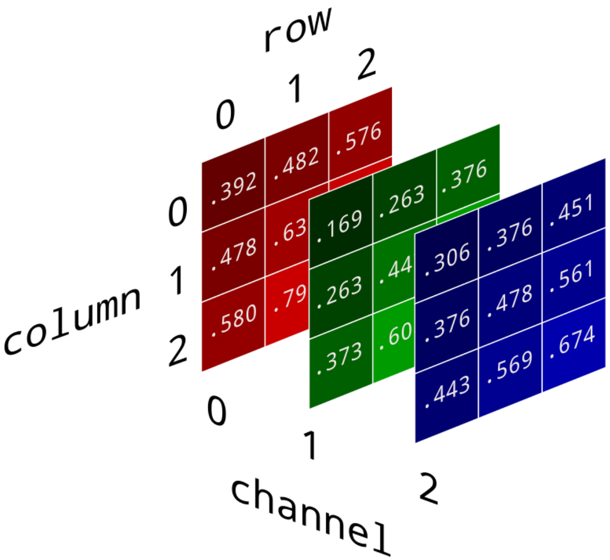
OpenCV Library

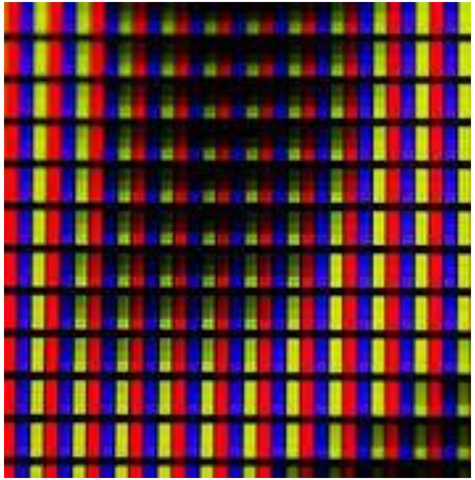
- Image manipulation
  - Read and display
  - Color space and channel manipulation
  - Resize, rescale, cropping
  - Transformations
  - Draw and Masking
- Object and Face detection
  - Contour, thresholding, and edge detection
  - Basic item detection and recognition
- Connecting with AI modules

[Open Computer Vision Library]

pip install opencv-python
Python >=3.6

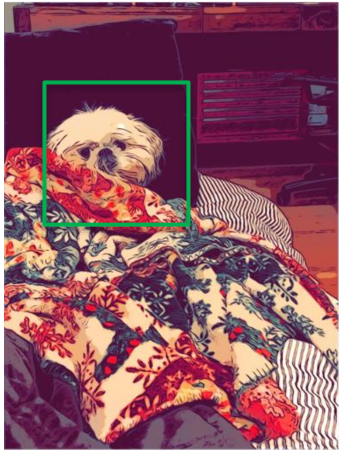
Image





https://brandonrohrer.com/convert_rgb_to_grayscale.html

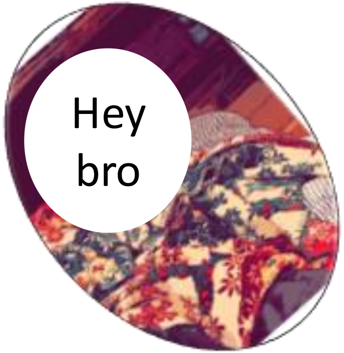
Functionalities



Img

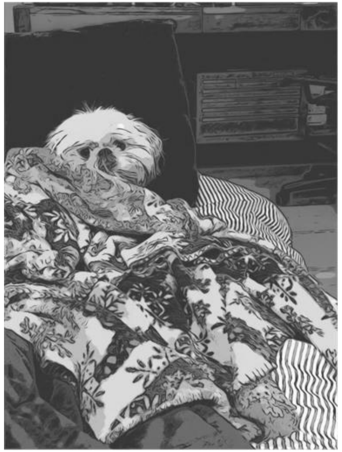
Resize

Crop



Mask & rotate

Transform



CVT

Contour / Edge

https://brandonrohrer.com/convert_rgb_to_grayscale.html

Code walkthrough